

Preserving Privacy in VR Telemetry Data

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ABSTRACT

Telemetry data is essential for optimizing VR systems and poses privacy risks due to re-identification vulnerabilities. Our study introduces a privacy-preserving framework using Wasserstein GANs with Gradient Penalty to generate synthetic datasets that retain the statistical integrity of real data while mitigating re-identification risks. Differentially Private Stochastic Gradient Descent is used to further enhance privacy by introducing controlled noise during model training. Our approach balances data utility with user privacy, enabling secure analysis of VR telemetry data for performance optimization, user experience improvements and potentially paving the way for safer and more ethical VR data practices.

Index Terms: Virtual-Reality Telemetry Data, User Privacy, Personally Identifiable Information, Synthetic Data Generation, Wasserstein GAN, Re-identification Risk, Privacy-Preserving Techniques

1 INTRODUCTION

In recent years, telemetry data has emerged as a cornerstone for advancing VR systems and applications. This data, encompassing head movements, hand gestures, and interaction patterns, gaze tracking, and physiological signals, enables researchers as well as developers to refine system performance, enhance user experiences, and implement adaptive VR application scenarios. Telemetry data also plays a crucial role in optimizing system performance by reducing latency and refining interaction precision, while facilitating adaptive systems that dynamically respond to user behavior. However, the detailed and behavioral specific nature of VR telemetry data presents a double-edged sword. While it provides unparalleled insights, it also exposes users to significant privacy risks.

The challenge lies in the uniqueness of VR telemetry data, which often contains identifiable behavioral patterns. Even with traditional anonymization methods, such as pseudonymization or data masking [5], these patterns still allow for re-identification of users. Advanced techniques will be required to safeguard user privacy without compromising the utility of the data.

To address these concerns, we introduce a synthetic data-generation framework leveraging Wasserstein Generative Adversarial Networks with Gradient Penalty [3] (WGAN-GP) and Differentially Private Stochastic Gradient Descent [1] (DP-SGD). The WGAN-GP model allows for the generation of high-fidelity synthetic datasets that mimic the statistical properties of real data, while DP-SGD ensures privacy by carefully introducing controlled noise during the training process to obscure the influence of individual data points. By applying these techniques to VR telemetry data, our work provides a novel solution for safeguarding privacy without compromising the utility of data, paving the way for safer and more ethical practices in immersive environments.

2 RELATED WORK

Significant prior research has highlighted the privacy risks inherent in VR telemetry data. Nair et al. [5] demonstrated the feasibility of user re-identification through head and hand movement data,

achieving high accuracy rates. Similarly, Miller et al. [4] revealed how behavioral patterns within VR telemetry could expose personal attributes, underscoring the limitations of traditional anonymization techniques.

To address these risks, advanced methods like differential privacy have been developed, which introduce controlled noise to limit the exposure of individual data points. Another approach is synthetic data generation, which creates datasets that mimic the statistical properties of real-world data, offering stronger privacy guarantees while preserving utility.

Generative Adversarial Networks [2] (GAN) have transformed the landscape of synthetic data generation. However, traditional GANs often face challenges like training instability, which are exacerbated in high-dimensional datasets such as VR telemetry data. To overcome these issues, Wasserstein GANs with Gradient Penalty [3] (WGAN-GP) enforce Lipschitz continuity via gradient penalties, thereby enhancing training stability. WGAN-GP has shown much promise in privacy-preserving applications by generating high-quality synthetic datasets that retain important data characteristics while at the same time reducing re-identification risks.

3 METHODOLOGY

Our study focuses on addressing privacy risks in VR telemetry data using a subset of the BOXRR-23 dataset [6], which includes telemetry records from 2,598 users. The methodology involves three key components: data preprocessing, synthetic data generation, and evaluation.

3.1 Data Preprocessing and Feature Selection

The dataset first was preprocessed to ensure consistency and eliminate irregularities that could impact model performance. Key telemetry attributes such as `saberDirX`, `saberDirY`, and `saberDirZ`, representing directional movements, were selected due to their importance in spatial interaction. Additional aggregated features, including `cutDirDeviation.mean`, `cutDistanceToCenter.std`, and `saberSpeed.mean`, were extracted to capture meaningful patterns.

All features were normalized using Scikit-learn's `StandardScaler` standardizing features to a mean of zero and unit variance. Outliers, defined as values beyond three standard deviations, were clipped to maintain statistical integrity of the dataset. Preprocessing ensured the data was suitable for training the WGAN-GP model and accurately represented real-world telemetry distributions.

To assess the privacy risks inherent in the data, a similarity-based re-identification process was conducted. Pairwise cosine similarity metrics were computed between telemetry feature vectors of real users to quantify re-identification risk. The analysis revealed a 100% identification accuracy, underscoring the need for robust privacy-preserving solutions.

3.2 Synthetic Data Generation with WGAN-GP

To mitigate privacy risks, WGAN-GP was employed to generate synthetic VR telemetry data. The model architecture was tailored to ensure stable training and high-quality data generation. The **Generator** transforms random noise into synthetic telemetry data through three fully connected layers:

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- **Layer 1:** 128 units, Batch Normalization, and Leaky ReLU activation ($\alpha = 0.2$).
- **Layer 2:** 256 units, Batch Normalization, and Leaky ReLU activation ($\alpha = 0.2$).
- **Output Layer:** Linear activation with units matching the feature dimensions of the dataset.

The **Critic** evaluates the quality of the generated data by estimating the Wasserstein distance between real and synthetic distributions. Its architecture includes:

- **Layer 1:** 256 units, Leaky ReLU activation ($\alpha = 0.2$), and Dropout (rate = 0.3).
- **Layer 2:** 128 units, Leaky ReLU activation ($\alpha = 0.2$), and Dropout (rate = 0.3).
- **Output Layer:** Single unit with linear activation.

To enforce Lipschitz continuity a **Gradient Penalty** was added to the critic's loss function ensuring stable training and preventing model collapse. The model's **Trainings Process** used the following procedure:

1. **Critic Updates:** For every generator update, the critic was updated five times to refine Wasserstein distance estimation.
2. **Generator Update:** The generator was optimized to produce data indistinguishable from real telemetry data.
3. **Differential Privacy Integration:** DP-SGD introduced noise to gradient updates, ensuring individual data privacy.

The training loop utilized a batch size of 64, a learning rate of 1×10^{-4} , and a latent dimension of 128.

3.3 Evaluation and Validation

The synthetic data was evaluated for both privacy preservation and data utility. Statistical similarity tests, including the Kolmogorov-Smirnov test and Kernel Density Estimation plots, were used to compare the distributions of real and synthetic data.

Re-identification risk assessments demonstrated a significant reduction in accuracy when synthetic data was used, dropping from 100% in real data to 0.04% with synthetic data, confirming the effectiveness of our approach. Additionally, visual analyses of feature distributions, as shown in figure 1, highlighted close alignment between real and synthetic datasets, further validating the fidelity of the generated data. Our comprehensive evaluation demonstrated that WGAN-GP effectively balances privacy and utility, making it a robust solution for privacy-preserving synthetic data generation in VR telemetry applications.

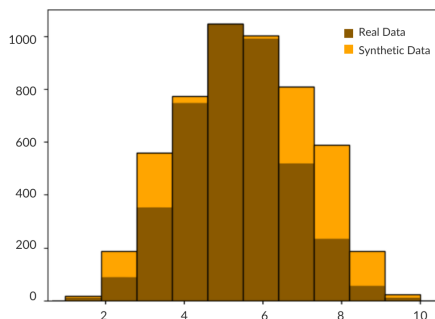


Figure 1: Comparison of skill score distributions for real and synthetic data. Both graphs follow a normal distribution with most users scoring between 4 and 7.

4 RESULTS AND LIMITATIONS

Our WGAN-GP-based framework effectively generates synthetic VR telemetry data that closely mirrors the statistical properties of real data while significantly reducing re-identification risks. The Kolmogorov-Smirnov test confirms that most features are well-replicated, with minor deviations in complex attributes, preserving the overall structure of the original dataset.

Figure 1 shows the distribution of skill scores (scaled between 1 and 10) for real and synthetic datasets, both exhibiting a roughly normal distributed behavior for most users with scores between 4 and 7. Synthetic data retains sufficient utility for analytical tasks, such as performance benchmarking and skill evaluation, while mitigating privacy concerns.

These results validate the effectiveness of WGAN-GP for privacy-preserving synthetic data generation while retaining analytical value for VR applications. While WGAN-GP was chosen for its stability with high-dimensional data future work will explore modern Transformer-based architectures to enhance data fidelity.

Despite promising results some limitations persist. Slight discrepancies were observed in the replication of complex features and the computational cost of training WGAN-GP on large datasets remains high. Addressing these challenges will improve efficiency and enable broader applications in VR telemetry and beyond.

5 CONCLUSIONS AND FUTURE WORK

Our study demonstrated the effectiveness of WGAN-GP-based framework in generating synthetic VR telemetry data that preserves the utility of real data while significantly reducing re-identification risks. The synthetic data closely mimics the statistical properties of original dataset, enabling analytical tasks such as performance benchmarking and skill evaluation without compromising user privacy.

We plan to enhance training efficiency and improve fidelity for complex attributes. Additionally, we are exploring real-time encryption methods to further strengthen privacy and expand the framework's applicability to domains like gaming and healthcare.

Our research offers a meaningful step to safeguard privacy in immersive digital experiences while fostering responsible VR innovation. By bridging the gap between privacy and utility, our work sets a foundation for secure and ethical practices in VR data analysis.

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